Final project

Replication and presentation of an empirical paper — 60% of the unit grade

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Overview

For the final project, your group will take one empirical paper from a list and, using your own code, reproduce some of its results. There are then two live parts: you **present the paper in class** and answer questions on it, and your **code is examined separately in a short Zoom meeting** with your group. There is no written report. The paper list will be released on Moodle on **Friday 31/07**.

Groups

You'll work in groups of 2 or 3, depending on final enrollment.

- By **Thursday 13/08**, email me the names of the people in your group.
- If you don't have a group, email me on the same day to let me know, and I'll pair you with a partner or add you to a group. I'll tell you who you're working with that same day.
- Either way, you'll come to class on **Tuesday 18/08** with your group and your ranked list of papers (explained in the next section).

Collaboration is within your group only: no collaborating with other groups on the code or the slides.

If problems arise within a group (workload, availability, disagreement), try to resolve them in the group first; if they persist, raise them with me well before the hand-in — not after grades are out. In documented cases of clearly unequal contribution I may adjust individual grades.

How papers are assigned

Each group gets a different paper. Here's how:

1. Before class on **Tuesday 18/08**, your group ranks all the papers on the list in order of preference — number 1 is the paper you'd most like to work on, down to the last one, the one you'd least like. This is your ranked list.
2. In class, each group will draw a random number. This sets the order in which groups pick.
3. We then go through the groups one at a time in that order. When it's your group's turn, you get the highest paper on your list that no earlier group has already taken.
4. Because papers get taken as we go, your group may not get its first choice — this is why you rank the full list, so there is always a next paper on it for you.
5. No two groups work on the same paper. Assignment happens in class that day.

What you need to do

Replication code

Each paper has its own document on Moodle, called `what_to_replicate` (released with the paper list on **31/07**): it sets out exactly what your code must produce — which figures and tables, from which data — and what is explicitly not required. Your job is to write code that produces every exhibit on your paper's list, preferably in Stata (contact me if you would like to use another software).

The coding is there for two reasons: it tests that you can put the tools from the lab sessions to work, and it makes you understand the paper's econometric analysis — and the data choices the authors made — at a level of detail that isn't always visible in the published paper itself.

For some papers, part of the job is building the working dataset yourselves from the raw source files provided: importing, cleaning and merging sources, and constructing variables. Where your paper's list asks for this, the choices involved are yours to make — document each one as a comment in your code.

Do not aim to reproduce the paper's exact numbers. Treat the published numbers as a benchmark, not as a target: you will not lose points simply because your results differ from the paper's. Where the data or software used leaves several reasonable ways to proceed (which source to use, which sample to keep, how to define a variable, which command to use), any decision you take is fine as long as the justification is sound — write the justification as a comment in your code, and be ready to defend it when

you're asked about your code. What costs points in your code is a coding mistake, not a defensible decision.

Some papers come with the authors' own replication package. I will make it available on Moodle when it exists. It is there for reference only: you are not expected to look at it, but you can if you want to. Either way, the code you hand in must be yours — written by you — and you must be able to defend it, line by line, in the code session.

Presentation (in class)

You'll present the paper in class in Weeks 11 and 12. Each group presents once, on one of four dates: **Tue 13/10** or **Thu 15/10** (Week 11), or **Tue 20/10** or **Thu 22/10** (Week 12). Each group has one 30-minute slot: a **20-minute presentation** followed by **10 minutes of questions**.

The presentation is about the paper, not about your replication. Present the paper's results — including results you were not asked to replicate — and use the paper's own figures and tables to do it. **Do not show figures or tables produced by your own code:** your code is discussed in the Zoom code session instead, so nothing produced by it belongs in your slides. You are expected to understand the whole paper, not just the parts your code reproduces.

The questions in class are about the paper: they can cover any part of it, whether or not you presented it and whether or not you replicated it. Questions about your code are asked in the code session, not in class.

You usually won't have time to show every result. Choose the ones that best capture what makes the paper important — what it adds compared to other work, and why it matters for policy.

The last slide of your presentation must be about policy: what we learn from the paper from a policy standpoint, and to what extent you think its results are directly applicable.

How you deliver the talk matters, and it is graded. Do not read. Reading your slides aloud, reading from a script, or reciting a memorised text will cost you marks, however good the content is. Speak to the audience: look at them, talk from your slides rather than through them, and use your own words. Brief glances at a note card to check you are on track are fine; a written-out talk read line by line is not. Your slides should be prompts for you and support for the audience, not a transcript — keep them light on

text so that reading them is not an option. This takes rehearsal: practise the talk out loud, as a group, in the time you'll actually have, before the day.

You can still be asked about results that are in the paper but that you didn't show. A good option is to keep some extra figures and tables in an appendix at the end of your slides for the parts of the paper you think are important but you don't have time to discuss, so you can turn to them if a question comes up.

All students are expected to attend every presentation, not only the day their own group presents. During the question period, the last few minutes will be used to put a couple of questions to the audience to check that everyone has followed the talk. Your answers as an audience member will not count towards your own grade — but they help measure how clearly the presenting group got their work across, and so may affect the presenting group's grade. It therefore matters that you attend and pay attention throughout.

Code session (on Zoom)

Your code is examined separately, in a short Zoom meeting with your group: **15 minutes** for a group of 2, **20 minutes** for a group of 3. There is nothing to prepare and nothing to present — we simply read the code you submitted.

- All group members attend the same meeting, and everyone is questioned.
- One of you shares their screen with the submitted code open. I choose which parts we look at.
- Questions are of this kind: what does this line do, why is it there, why did you do it this way rather than another, and what does this choice imply for the result.
- **No live reruns:** you won't be asked to run the code, fix anything, or write anything during the meeting, and the code is not edited during the session — it is the version you submitted.

Sessions are held after the submission deadline, between **Tuesday 13/10** and **Friday 23/10**. Once the code is in, I'll post a booking sheet on Moodle and each group signs up for a slot. If nobody in the group can make any of the slots, email me before the sheet closes.

The code session is not recorded and there is no slide deck for it — bring nothing but your code and your ability to explain it.

What to hand in — due Monday 12/10 (upload to Moodle)

- Your slides.
- Your code, producing all the exhibits listed in your paper's `what_to_replicate` document.
- No written report.

Late submissions lose 25% of the project mark. Hard cutoff: nothing is accepted after **9am on Tuesday 13/10**, when the presentations begin — I review everyone's code and slides before the talks, so past that point your group is assessed on whatever has been submitted.

Slides and code are submitted through Moodle and run through a similarity-detection system, which compares them with online sources, published work, the authors' code, and other students' submissions — past and present.

One member uploads on behalf of the group, having checked that everyone agrees it is the final version; all members are jointly responsible for the submission being correct and on time.

Data and code

For each paper you'll be given the paper itself (with its online appendix where relevant), the data needed to produce the required exhibits — for some papers, raw source files you assemble yourselves — and READMEs describing everything in the folder.

Using AI

- You may use AI to help you understand the paper, build your slides, and write your code.
- You may not use AI during the presentation, and you may not use AI during the code session — no AI window open, no prompting between questions.
- You must understand everything in your slides and every line of your code. In class, I may ask you to explain a specific result; in the code session, I may ask you to explain a specific line. You will also need to justify any coding decision taken.

How you'll be graded

The project counts for **60%** of your final grade in this unit.

Your project grade is made of **four parts**. To keep things simple, every percentage below is a share of the total project grade: the four parts add up to 100%, and there are no percentages of percentages to work out.

1. The submitted code — 35% — group grade (shared by everyone in the group)

Whether your code runs and produces every exhibit on your paper's list, is free of coding mistakes, and is clean and well organised, with your choices documented and justified in comments. How close your numbers come to the paper's published ones is not part of the grade (see "Replication code" above).

2. The code session — 15% — individual grade (your own)

How you personally answer questions on the code in the Zoom session — what a specific line does, why it is there, and why you made the choices your comments document. Each group member is expected to be across all of the replication code, not only the part they wrote. Not attending the code session means no mark for this part.

3. The slides and the presentation as a whole — 35% — group grade (shared by everyone in the group)

It looks at the following (not in order of importance):

- **The method** — how clearly your slides and talk show that the group understands the approach the paper uses and the assumptions it relies on to reach a causal conclusion.
- **Applying it in the real world** — how well the group addresses the policy insights that can be drawn from the paper.
- **The presentation as a whole** — your choice of which results to show, how the talk is structured, staying within time, and how well the audience follows (judged partly by how well audience members can answer questions about your work).

4. Your own performance in class — 15% — individual grade (your own), in two equal halves

- **Delivering your part of the talk (7.5%)** — how clearly and effectively you present the portion you're responsible for. Delivery counts, not just content: whether you speak to the audience rather than read to them, whether you are clear and audible, and whether you keep the room with you. Reading your slides or a script aloud will lower this mark. This is judged on your presentation skill relative to the difficulty of your part: explaining a harder section is not held to a higher bar than explaining an easier one, so whoever presents the more technical material is not at a disadvantage. Every group member is expected to present a part, and you're free to divide the talk between you however you see fit.
- **Answering questions on the paper, in class (7.5%)** — you can be asked about any part of it: the question it answers, the identification strategy and the assumptions it rests on, any result (including ones you did not replicate), and its limitations. Each group member is expected to be across all of the paper.

The same four numbers, read two ways:

- **By what is graded:** the code side (submitted code 35% + code session 15%) counts for **50%**, and the presentation side (slides and presentation 35% + your performance in class 15%) counts for the other **50%**.
- **By who earns it:** group work (35% + 35%) counts for **70%**, and your individual live performance (15% + 15%) for **30%**.

Part	Graded as	Share of project grade
1. Submitted code	Group	35%
2. Code session (Zoom)	Individual	15%
3. Slides and presentation as a whole	Group	35%
4. Delivery (7.5%) + questions in class (7.5%)	Individual	15%
Total		**100%**

Grading also takes into account how hard the paper's methods and results are to reproduce, and how many people are in your group.

Feedback

You'll get oral feedback during the question period and during the code session; your marks and short written comments within 10 working days of your code session; and your final unit result at the official results release.